

Sensitivity test - Window type

[ODSA_README](#)

 [ODSA-AIM_and_METHOD](#)

The current setup

Fixed 50 km **rectangular window with 10 km step**. The Lomb-Scargle PSD is computed per-window, then averaged. Peaks are detected in the whitened residuals after power-law fitting.

- ✓ Test 3 window types (e.g., Rectangular Hann, Tuckey)
 - ~~Hann is the most aggressive taper — best sidelobe suppression but you lose the most signal amplitude at the edges.~~
 - ~~Tukey with alpha=0.5 tapers the outer 25% on each side while leaving the central 50% untouched. This is gentler than Hann (which tapers everything).~~

WHY

The rectangular window may produce side lobes that could create spurious peaks in the whitened residuals. *(Since I'm fitting a power law across a broad frequency band and then looking for peaks in the whitened residuals, spectral leakage from the rectangular window could create spurious peaks at harmonics of strong signals.)* A different window type might give cleaner peak detection without substantially compromising frequency resolution.

Findings summary

Metric	Rectangular	Hann	Tuckey
Trajectories	20	20	20
Segments	38	38	38
β exponent	2.2, [1.3, 2.8]	2.3, [1.7, 3.0]	2.3, [1.4, 2.9]
ζ exponent	0.6, [0.2, 0.9]	0.7, [0.3, 1.0]	0.7, [0.2, 1.0]
Vertical Relief	1311.3 m	1311.3 m	1311.3 m
Bed elevation range	[-1874.8, 1225.2] m	[-1874.8, 1225.2] m	[-1874.8, 1225.2] m
Ice thickness	2990.6 m	2990.6 m	2990.6 m
flow orientation	18 $\parallel$ /14 $\perp$ /6 O	18 $\parallel$ /14 $\perp$ /6 O	18 $\parallel$ /14 $\perp$ /6 O
wavelength	~	~	~
candidates	4, [10777, 13772]	5, [8441, 19356]	5, [10416, 16815]
confirmed count	1456	1225	1420
confirmed mean	1068.5 m	744.3 m	913.4 m

Metric	Rectangular	Hann	Tuckey
confirmed range	[54.2, 22072.0] m	[54.2, 18836.5] m	[54.2, 33400.8] m
exported windows	229	229	229

The window type actually shifts some metrics:

1. Roughness exponents increase slightly, the taper of the edges suppresses edge energy leak.
2. Confirmed wavelength count drops, the taper removes some peaks that were likely sidelobe artifacts.
3. Candidate wavelength counts went from 4 to 5 maybe spectral leakage was masking real long-wavelength peak.

	Rectangular	Hann	Tuckey
Segment fits			
Linear regression	slope=0.0008 R ² =0.005 p=6.59e-01	slope=0.0015 R ² =0.019 p=4.04e-01	slope=0.0001 R ² =0.000 p=9.78e-01
cos ² θ	$\beta_{\parallel} = 2.126 \pm 0.073$ $\beta_{\perp} = 2.193 \pm 0.081$ $\Delta\beta = -0.067$ R ² =0.0081	$\beta_{\parallel} = 2.36 \pm 0.08$ $\beta_{\perp} = 2.25 \pm 0.08$ $\Delta\beta = +0.11$ R ² = 0.019	$\beta_{\parallel} = 2.31 \pm 0.08$ $\beta_{\perp} = 2.32 \pm 0.09$ $\Delta\beta = -0.00$ R ² =0.000
Window fit			
Linear regression	slope=0.0003 R ² =0.0009 p=6.4876e-01	slope=-0.0005 R ² =0.0027 p=4.3379e-01	slope=-0.0001 R ² =0.0003 p=8.1177e-01

The **Tukey window**, as the gentler middle ground between rectangular and Hann, produces results closest to the rectangular case while still providing some sidelobe suppression. All three window types converge on the same classification

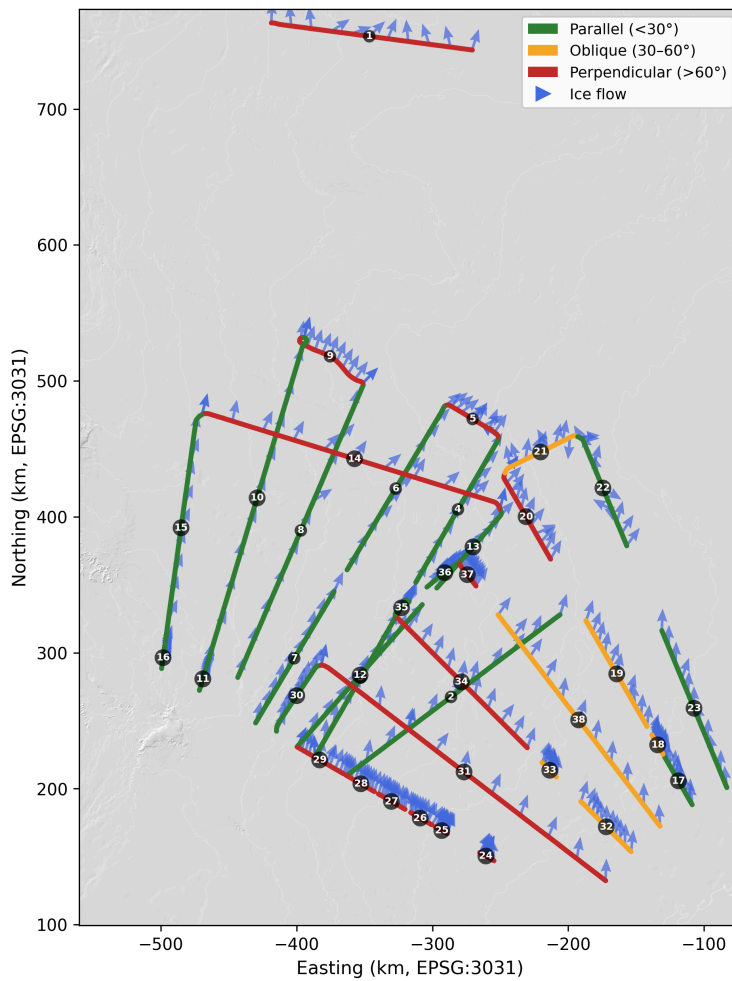
TESTING DATA SET: Pensacola Pole Basin (PPB)

Radar tracks: BAS_2015_POLARGAP_AIR_BM3

Radar Flight Tracks - Antarctica Overview



Ice Flow Orientation: POLARGAP_2015_Fig1_Pensacola_Pole
P32.6



ODSA (**stratified**) measured roughness of each flight line track by orientation
The majority of segments were found to be parallel (n=18) and perpendicular (n=14).
while only 6 segments were found to be oblique.

RESULTS

Rectangular window

Terminal

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=== POLARGAP_2015_Fig1_Pensacola_Pole SUMMARY ===
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RESULTS SUMMARY (20 trajectories aggregated)
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VERTICAL RELIEF (Max-Min):
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-> Mean: 1311.3m | Range: [442.2, 2437.3]m
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ABSOLUTE BED ELEVATION:
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-> Min: -1874.8m | Max: 1225.2m (across all trajectories)
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```
AVG SEGMENT LENGTH:
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```
-> Mean: 114066.9m | Range: [20970.2, 262973.5]m
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POWER LAW EXPONENT (Beta):
-> Mean: 2.2 | Range: [1.3, 2.8]
HURST EXPONENT:
-> Mean: 0.6 | Range: [0.2, 0.9]
MEAN ICE THICKNESS:
-> Mean: 2990.6m | Range: [1801.3, 3821.0]m
.....
FLOW ORIENTATION (All Segments):
-> Oblique: 6 segments (15.8%)
-> Perpendicular: 14 segments (36.8%)
-> Parallel: 18 segments (47.4%)
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
-> Count: 1456
-> Mean: 1068.5m | Range: [54.2, 22072.0]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
-> Count: 4
-> Range: [10777m, 13772m]
.....
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Exported 229 window rows and 38 segment rows

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Observations

IFPA categorised the region containing these radar tracks as **Alpine (subglacial) / Selective erosion**. While the ODSA results indicate this region has a mean β value of 2.2 with a large range: [1.3, 2.8]. Putting the result between the **hard bed** ("Brownian": complex and rough) and the **sedimentary Infill** category ($\beta > 2.5$). This means that ODSA would classify this area as a combination of high relief terrain with some smoother areas.

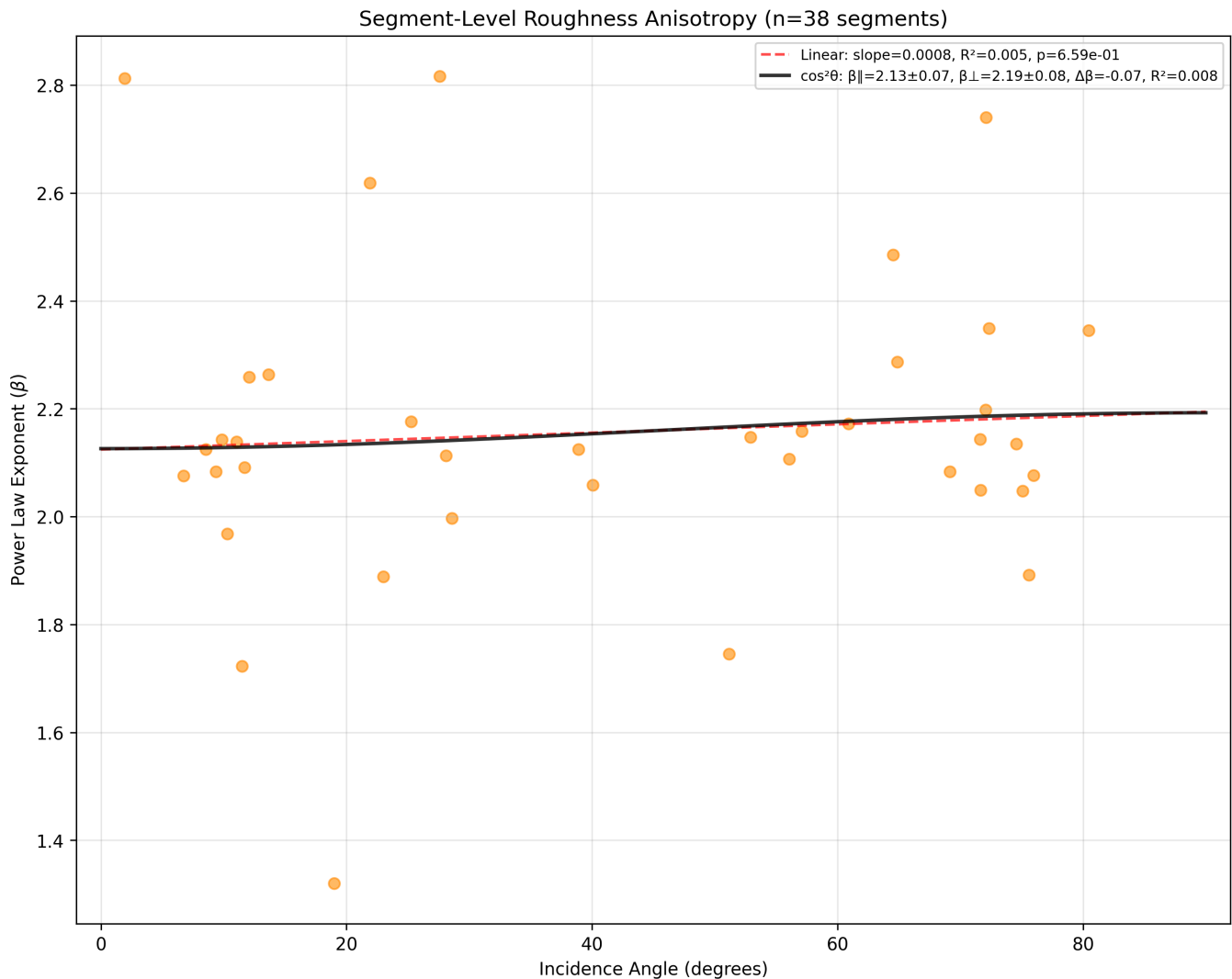
The mean value for the vertical relief of 1311.3 m | Range: [442.2, 2437.3]m

The height difference between the highest and lowest points available in the data for this area are Min: -1874.8 m | Max: 1225.2 m (across all trajectories).

The mean confirmed wavelength of 1068.5 m with the range going up to ~22 km indicates the roughness is dominated by structures at roughly the km-scale.

Anisotropy

Segment-level Roughness FIT



The $\cos^2\theta$ fit: $\beta_{\parallel} = 2.13 \pm 0.07$, $\beta_{\perp} = 2.19 \pm 0.08$, $\Delta\beta = -0.07$, $R^2=0.008$

The error bars of the parallel and perpendicular beta parameters **overlap** substantially:

2.13 ± 0.07 gives [2.06, 2.20], and 2.19 ± 0.08 gives [2.11, 2.27], this means that parallel and perpendicular roughness are indistinguishable

The anisotropy signal $\Delta\beta = -0.07$

Its significance can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.07}{\sqrt{0.07^2 + 0.08^2}} \approx \frac{0.07}{0.106} \approx 0.66$

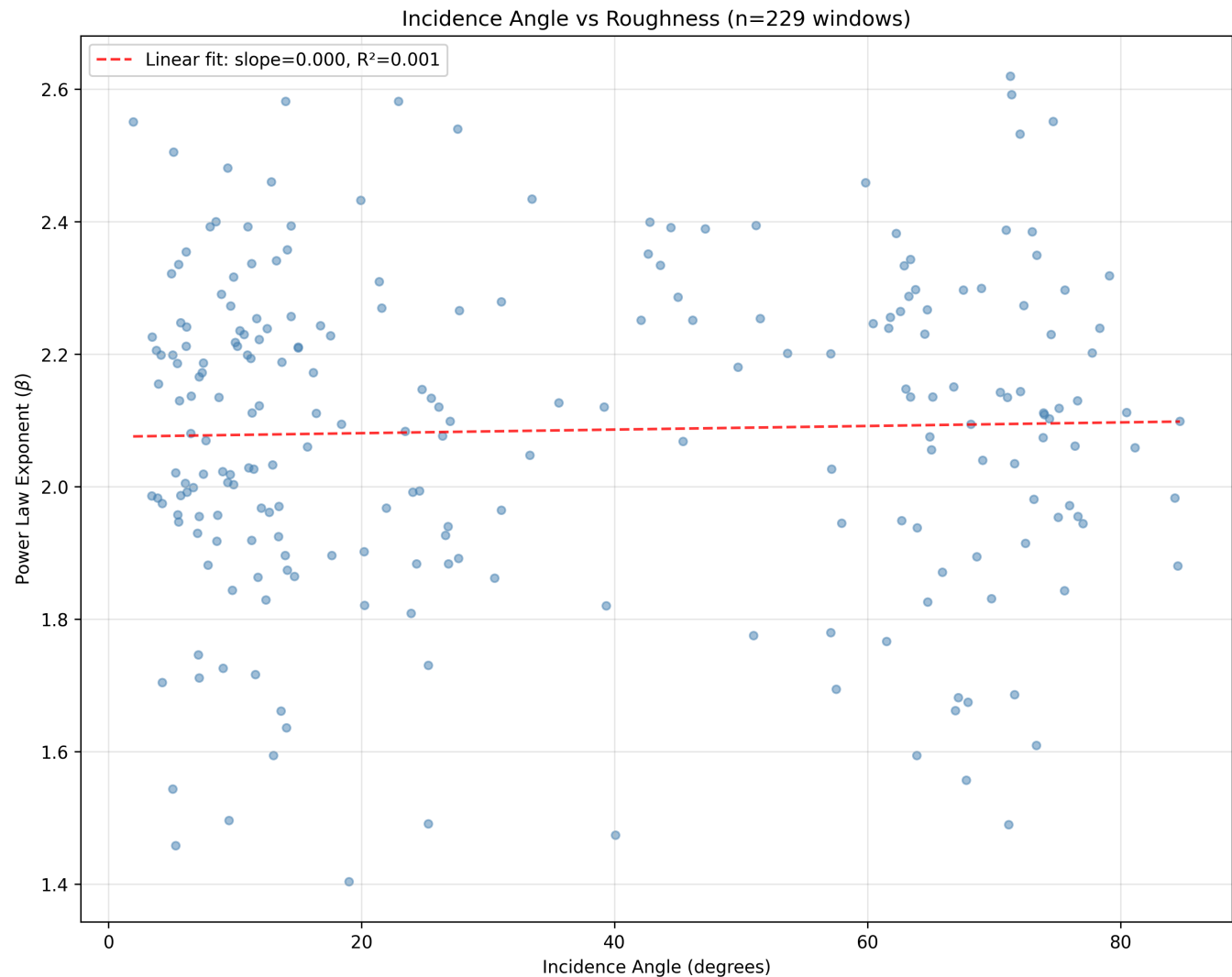
This is below 1.96 (95% confidence interval), which means that the result is not significant.

The $R^2=0.008$ implies that the model explains only 0.8% of variance. **Essentially no directional dependence detected.** This provides support for the **alpine** landscape classification (**isotropic landscape**) and rules out **selective erosion**.

The linear fit at the segment level has **slope=0.0008**, **$R^2=0.005$** , **$p=6.59e-01$**

Since **$p < 0.05$** means significant. At 0.659 this result is non-significant (there's a 66% chance you'd see this slope (or steeper) from pure noise). This means that the slope is essentially zero.

Window-level Roughness FIT



The window level analysis show a near-flat line fit (small slope, very modest R^2). That means that there is no systematic dependence of roughness on flow orientation, this is consistent with a non-streamlined landscape providing support for the **Alpine landscape** classification.

Conclusion:

PPB appears to be an Alpine region based on its isotropy and its bedrock hardness characteristics.

Hann window

Terminal

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==== POLARGAP_2015_Fig1_Pensacola_Pole SUMMARY ====
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RESULTS SUMMARY (20 trajectories aggregated)
-----
VERTICAL RELIEF (Max-Min):
-> Mean: 1311.3m | Range: [442.2, 2437.3]m
```

```

ABSOLUTE BED ELEVATION:
-> Min: -1874.8m | Max: 1225.2m (across all trajectories)
AVG SEGMENT LENGTH:
-> Mean: 114066.9m | Range: [20970.2, 262973.5]m
POWER LAW EXPONENT (Beta):
-> Mean: 2.3 | Range: [1.7, 3.0]
HURST EXPONENT:
-> Mean: 0.7 | Range: [0.3, 1.0]
MEAN ICE THICKNESS:
-> Mean: 2990.6m | Range: [1801.3, 3821.0]m
.....
FLOW ORIENTATION (All Segments):
-> Parallel: 18 segments (47.4%)
-> Oblique: 6 segments (15.8%)
-> Perpendicular: 14 segments (36.8%)
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
-> Count: 1225
-> Mean: 744.3m | Range: [54.2, 18836.5]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
-> Count: 5
-> Range: [8441m, 19356m]
.....
=====
Exported 229 window rows and 38 segment rows

```

Observations

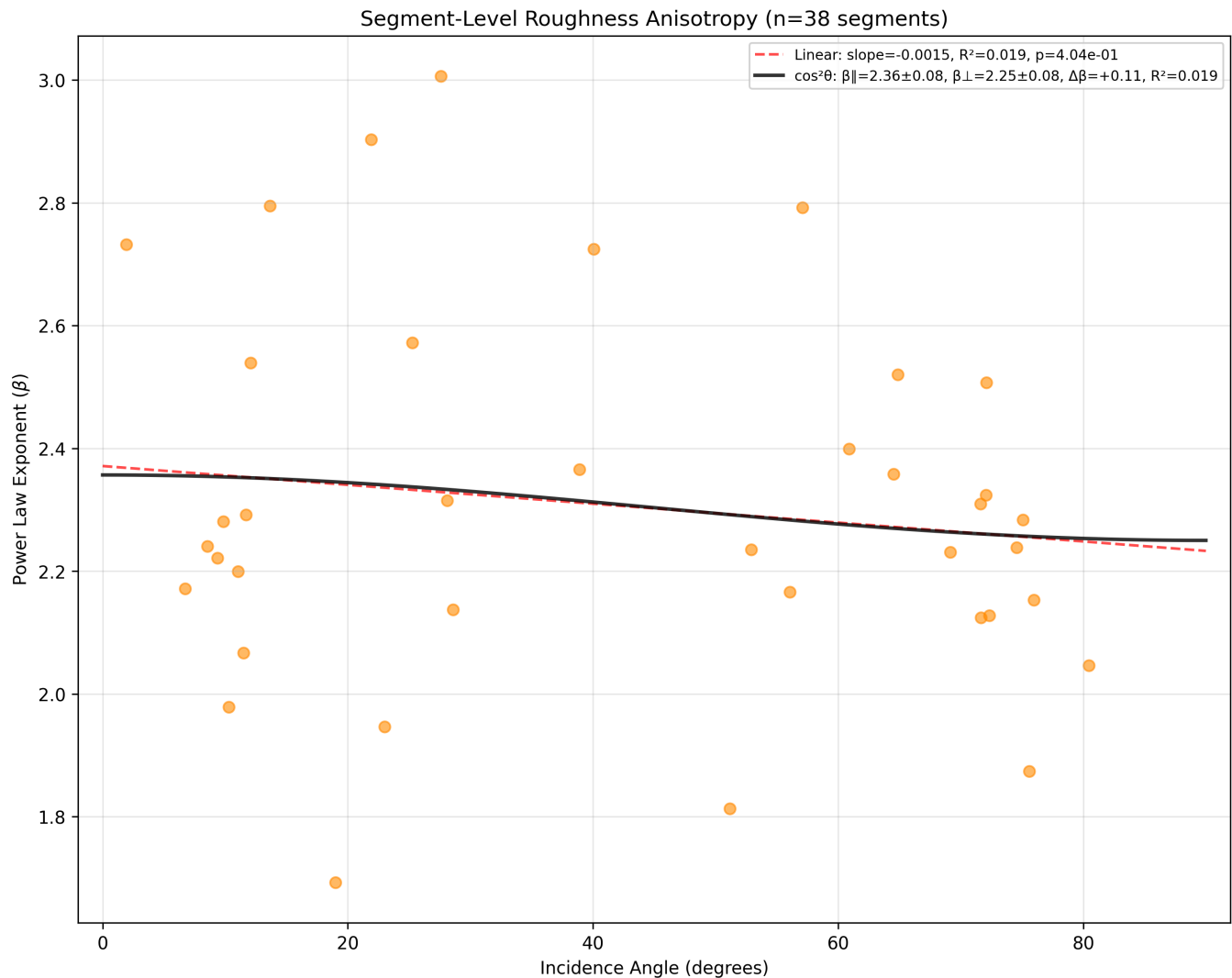
Mean vertical relief and range, aggregated height minimum and maximum values are all IDENTICAL to the rectangular window condition.

The mean $\beta = 2.3$ and the range is [1.7, 3.0], an increase from the rectangular window case.

The mean confirmed wavelength of 744.3 m with the range going up to ~19 km indicates the roughness is dominated by structures at roughly the sub-km-scale.

Anisotropy

Segment-level Roughness FIT



The $\cos^2\theta$ fit: $\beta_{\parallel} = 2.36 \pm 0.08$, $\beta_{\perp} = 2.25 \pm 0.08$, $\Delta\beta = +0.11$, $R^2=0.019$

The error bars of the parallel and perpendicular beta parameters **overlap** moderately:

2.36 ± 0.08 gives [2.28, 2.44], and 2.25 ± 0.08 gives [2.17, 2.33], this means that parallel and perpendicular roughness are not clearly distinguishable, though the overlap is narrower than in the rectangular and Tukey cases.

The anisotropy signal $\Delta\beta = +0.11$

Its significance can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.11}{\sqrt{0.08^2 + 0.08^2}} \approx \frac{0.11}{0.113} \approx 0.97$

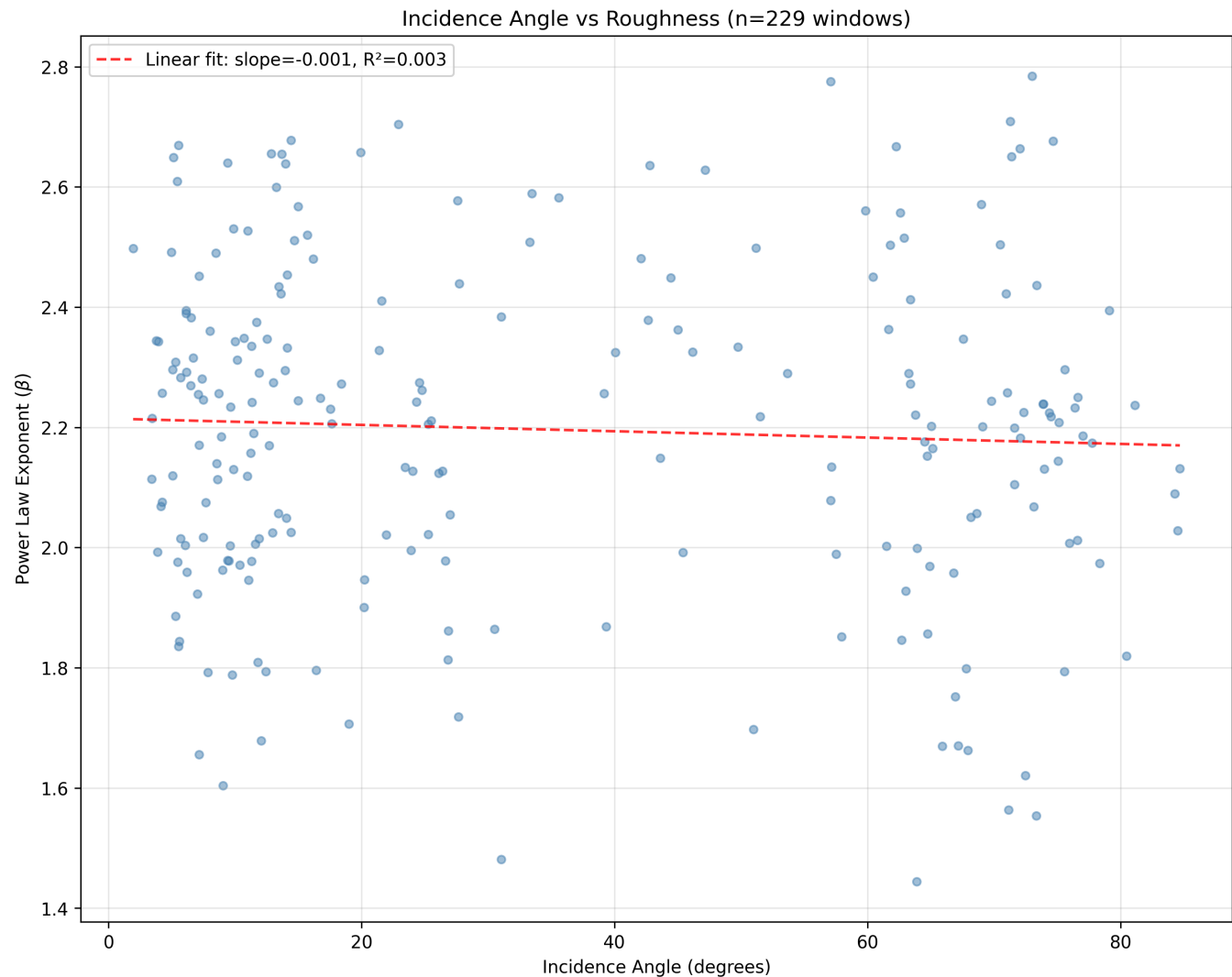
This is below 1.96 (95% confidence interval), which means that the result is not significant.

The $R^2=0.019$ implies that the model explains only 1.9% of variance. **Essentially no directional dependence detected.** This provides support for the **alpine** landscape classification (**isotropic landscape**) and rules out **selective erosion**.

The linear fit at the segment level has **slope=0.0015**, **$R^2=0.019$** , **$p=4.04e-01$**

Since **$p < 0.05$** means significant. At 0.404 this result is non-significant (there's a 40% chance you'd see this slope (or steeper) from pure noise). This means that the slope is essentially zero.

Window-level Roughness FIT



The window level analysis show a near-flat line fit (small slope, very modest R^2). That means that there is no systematic dependence of roughness on flow orientation, this is consistent with a non-streamlined landscape providing support for the **Alpine landscape** classification.

Conclusion:

The Hann window shows a marginally larger slope $\Delta\beta = +0.11$ compared to rectangular (-0.07) and Tukey (-0.00), but this is not statistically significant ($z \approx 0.97$). The slight increase in apparent anisotropy is likely an artifact of the aggressive tapering amplifying noise in the fewer remaining peaks, rather than a real physical signal. The classification remains consistent across all three window types.

Tuckey window

Terminal

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==== POLARGAP_2015_Fig1_Pensacola_Pole SUMMARY ====
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RESULTS SUMMARY (20 trajectories aggregated)
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VERTICAL RELIEF (Max-Min):
  -> Mean: 1311.3m | Range: [442.2, 2437.3]m
ABSOLUTE BED ELEVATION:
  -> Min: -1874.8m | Max: 1225.2m (across all trajectories)
AVG SEGMENT LENGTH:
  -> Mean: 114066.9m | Range: [20970.2, 262973.5]m
POWER LAW EXPONENT (Beta):
  -> Mean: 2.3 | Range: [1.4, 2.9]
HURST EXPONENT:
  -> Mean: 0.7 | Range: [0.2, 1.0]
MEAN ICE THICKNESS:
  -> Mean: 2990.6m | Range: [1801.3, 3821.0]m

.....
FLOW ORIENTATION (All Segments):
  -> Perpendicular: 14 segments (36.8%)
  -> Oblique: 6 segments (15.8%)
  -> Parallel: 18 segments (47.4%)

.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
  -> Count: 1420
  -> Mean: 913.4m | Range: [54.2, 33400.8]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
  -> Count: 5
  -> Range: [10416m, 16815m]

.....
=====
Exported 229 window rows and 38 segment rows

```

Observations

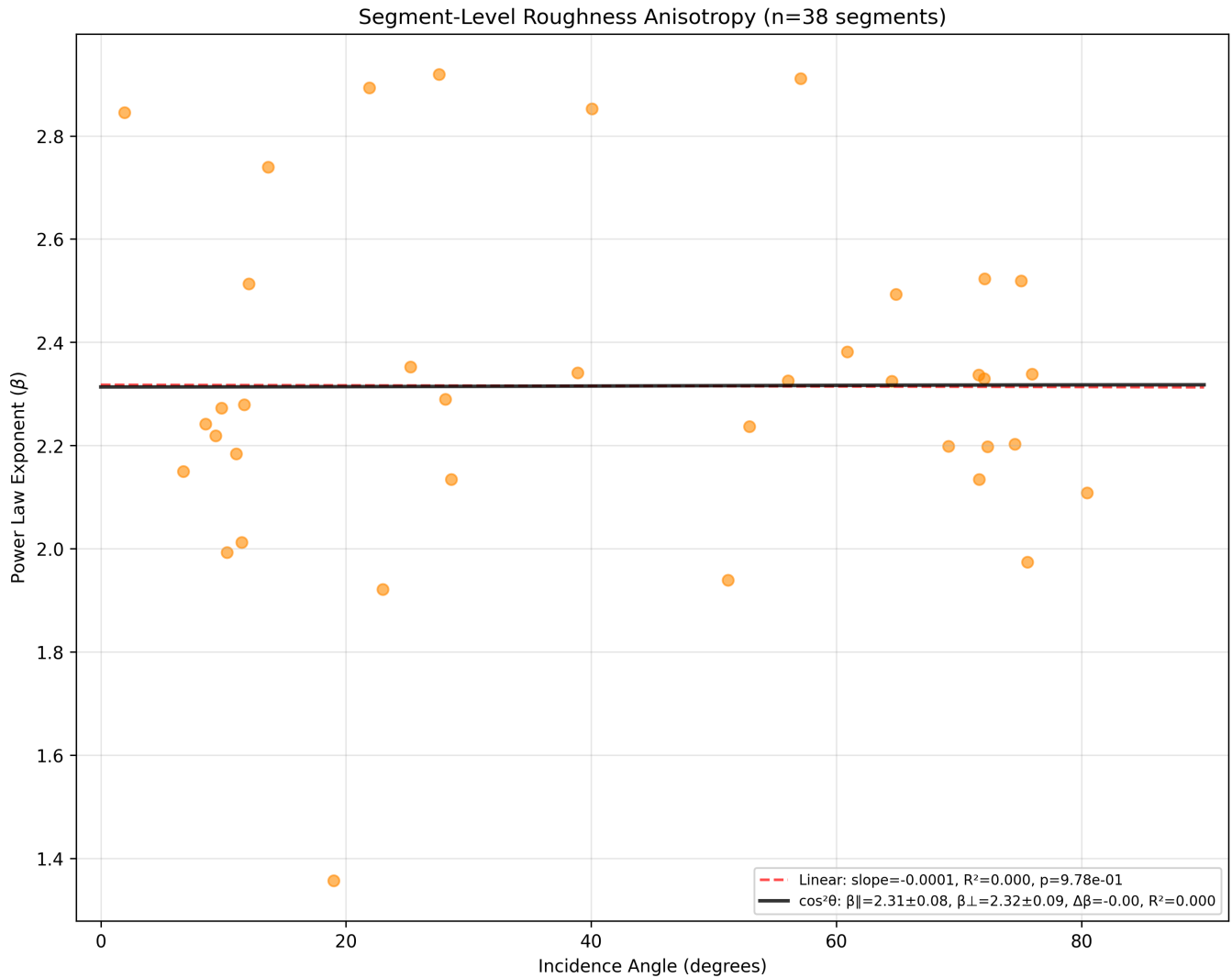
Mean vertical relief and range, aggregated height minimum and maximum values are all IDENTICAL to the rectangular window condition.

The mean $\beta = 2.3$ and the range is [1.4, 2.9], an increase from the rectangular window case.

The mean confirmed wavelength of 913.4 m with the range going up to ~33 km indicates the roughness is dominated by structures at roughly the sub-km-scale.

Anisotropy

Segment-level Roughness FIT



The $\cos^2 \theta$ fit: $\beta_{\parallel} = 2.31 \pm 0.08$, $\beta_{\perp} = 2.32 \pm 0.09$, $\Delta\beta = -0.00$, $R^2=0.000$

The error bars of the parallel and perpendicular beta parameters **overlap** completely:

2.31 ± 0.08 gives [2.23, 2.39], and 2.32 ± 0.09 gives [2.23, 2.41], this means that parallel and perpendicular roughness are indistinguishable.

The anisotropy signal $\Delta\beta = -0.00$

Its significance can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.00}{\sqrt{0.08^2 + 0.09^2}} \approx 0$

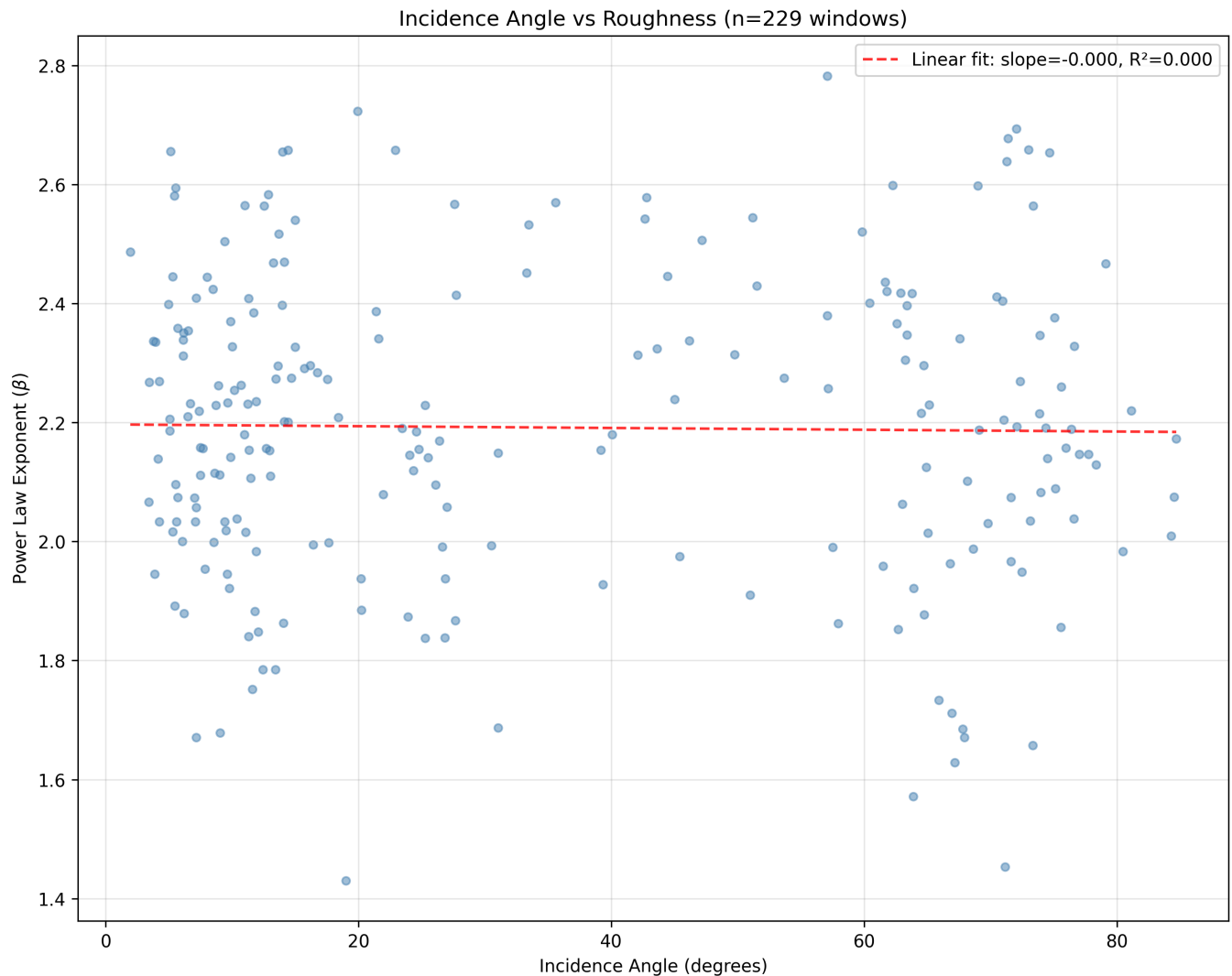
This is below 1.96 (95% confidence interval), which means that the result is not significant.

The $R^2=0.000$ implies that the model explains 0% of variance. **Essentially no directional dependence detected.** This provides support for the **alpine** landscape classification (**isotropic landscape**) and rules out **selective erosion**.

The linear fit at the segment level has **slope=0.0001**, **$R^2=0.000$** , **$p=9.78e-01$**

Since **$p < 0.05$** means significant. At 0.978 this result is non-significant (there's a 98% chance you'd see this slope (or steeper) from pure noise). This means that the slope is essentially zero.

Window-level Roughness FIT



The window level analysis show a near-flat line fit (zero slope, $R^2 = 0$). That means that there is no systematic dependence of roughness on flow orientation, this is consistent with a non-streamlined landscape providing support for the **Alpine landscape** classification.

Conclusion:

The Tukey window produces nearly identical anisotropy results to the rectangular case, with $\Delta\beta \approx 0.00$ and $R^2 = 0.000$, confirming that the window type does not introduce spurious directional signals.